

VERA RBL-XE

FPGA Build & Programming Guide

iCE40UP5K – Rev 3

6502 bus controller with RAMbo 256 KB, PBI RAM/ROM,
and external SRAM support

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1 Overview

This document describes the complete workflow for building and programming the FPGA firmware for the VERA RBL-XE module. The firmware implements a 6502 bus controller on a Lattice iCE40UP5K, replacing the ESP32-S3 MCU used in the previous design.

1.1 Architecture summary

- **PBI RAM** (\$D600–\$D7FF, 512 B): stored in iCE40 BRAM (1 EBR block).
- **PBI ROM** (\$D800–\$DFFF, 2 KB): stored in iCE40 BRAM (4 EBR blocks), pre-loaded at synthesis time from the assembled 6502 driver binary.
- **RAMbo** (\$4000–\$7FFF, 16 KB window, 256 KB total): served by an external 512K×8 SRAM (e.g. AS6C4008-55SIN). The FPGA controls chip-enable, output-enable, write-enable, and the upper four address bits (bank), using only 7 new I/O pins.
- **PHI2 synchronisation**: PHI2 is treated as a data input and synchronised to the 25 MHz FPGA clock via a two-stage flip-flop chain, producing a single-cycle `phi2_rise` pulse used to trigger all registered logic.
- **Two bitstreams – VERA_IS_PBI generic**: the top-level entity exposes a VHDL-2008 boolean generic that selects the bus interface at synthesis time. *PBI mode* (`VERA_IS_PBI=true`): latch at \$D1FF, PBI RAM and ROM in BRAM (5 EBR blocks), EXTSEL for PBI RAM and RAMbo, MPD for PBI ROM reads. *CCTL mode* (`VERA_IS_PBI=false`): latch at \$D5FF (write \$80 to select, \$00 to deselect), no BRAM (Yosys eliminates dead code: 0 EBR), EXTSEL for RAMbo only, MPD always inactive.
- **Hardware configuration pins**: `PBI_ID_SEL` (pin 27) is wired via PCB jumper to one of D[0..7]; the sampled bit value latches `VERA_CS` in PBI mode (`vera_cs_latch <= not PBI_ID_SEL`). `RAMBO_EN` (pin 28) with pullup (VCC) enables RAMbo; with pulldown (GND) disables it regardless of `PBCTL[2]`.

1.2 Memory map

Range	Size	Function	Implementation
\$4000–\$7FFF	16 KB window	RAMbo (256 KB banked)	External SRAM
\$D1FF	1 B	VERA_CS latch (write)	Register
\$D301	1 B	PIA PORTB snoop	Register
\$D303	1 B	PIA PBCTL snoop	Register
\$D600–\$D7FF	512 B	PBI RAM	BRAM (1 EBR)
\$D800–\$DFFF	2 KB	PBI ROM	BRAM (4 EBR)

Note (CCTL mode): when built with `VERA_IS_PBI=false` the `VERA_CS` latch address changes to \$D5FF (write \$80 to select, \$00 to deselect), and PBI RAM/ROM are absent (0 EBR).

2 Prerequisites

2.1 FPGA toolchain

All tools below are open-source. Install on Debian/Ubuntu:

```
sudo apt install fpga-icestorm nextpnr-ice40 yosys ghdl
```

Note: the `ghdl-yosys-plugin` is required for VHDL synthesis and is generally not available as a package. Build it from source:

```
git clone https://github.com/ghdl/ghdl-yosys-plugin
```

```
cd ghdl-yosys-plugin
make
sudo make install
```

2.2 Flash programming tools

```
# openFPGALoader -- recommended, supports many programmers
sudo apt install openFPGALoader

# iceprog is already included in fpga-icestorm
```

2.3 6502 assembler (cc65)

The PBI ROM driver is written in 6502 assembly and built with the cc65 toolchain.

```
sudo apt install cc65
```

2.4 Python 3

Required to run `gen_pbi_rom_pkg.py`, which converts the assembled 6502 binary into a VHDL constant package.

```
sudo apt install python3
```

2.5 Tool roles at a glance

Tool	Input	Output	Purpose
ca65 / ld65	.s sources	.bin	6502 assembly
gen_pbi_rom_pkg.py	.bin	.vhd	ROM initialisation
ghdl (plugin)	VHDL sources	RTLIL	VHDL elaboration
yosys	RTLIL	.json	Synthesis
nextpnr-ice40	.json + .pcf	.asc	Place & route
icepack	.asc	.bin	Bitstream packing
openFPGALoader	.bin	SPI flash	Programming

3 Build Flow

3.1 Step 1 – Assemble the 6502 ROM driver

The PBI ROM at \$D800–\$DFFF contains a 2 KB 6502 driver assembled from `firmware/MCU/6502/src/vera_pbi_`

```
cd firmware/MCU/6502
make
```

Listing 1: Assemble the 6502 driver

This single command runs three sub-steps:

1. **Assemble:** ca65 compiles `vera_pbi_handler.s` into `vera_pbi_handler.o`.
2. **Link:** ld65 links with `pbi-driver.ld` to produce `vera_pbi_handler.bin`.
3. **Generate VHDL ROM package:** `gen_pbi_rom_pkg.py` reads the binary and writes `../../FPGA/pbi_rom_pkg.vhd` with the `PBI_ROM_INIT` constant.

To regenerate only the VHDL package when the binary already exists:

```
make fpga-rom
```

3.2 Step 2 – Synthesis

The top-level entity exposes a VHDL-2008 boolean generic, `VERA_IS_PBI` (default `true`), that selects the bus interface at synthesis time. The Makefile invokes Yosys twice, producing two independent bitstreams from the same source:

```
cd firmware/FPGA
yosys -m ghdl -p \
    "ghdl --std=08 -gVERA_IS_PBI=true \
        custom_types.vhd pbi_rom_pkg.vhd C6502_Interface.vhd \
        -e C6502_Interface; \
    synth_ice40 -top C6502_Interface -json C6502_Interface_PBI.json"
```

Listing 2: PBI synthesis (`VERA_IS_PBI=true`)

```
yosys -m ghdl -p \
    "ghdl --std=08 -gVERA_IS_PBI=false \
        custom_types.vhd pbi_rom_pkg.vhd C6502_Interface.vhd \
        -e C6502_Interface; \
    synth_ice40 -top C6502_Interface -json C6502_Interface_CCTL.json"
```

Listing 3: CCTL synthesis (`VERA_IS_PBI=false`)

The `-m ghdl` flag loads the GHDL plugin into Yosys, which handles VHDL analysis and elaboration. When `VERA_IS_PBI=false`, Yosys eliminates all PBI RAM/ROM dead code, reducing resource usage dramatically. Measured values after place-and-route with `nextpnr-ice40`:

Resource	PBI	CCTL	Available	% (PBI)
ICESTORM_LC (total)	87	1	5280	1%
of which LUT4 only	52	1		
of which LUT4+DFF	5	0		
of which DFF only	5	0		
of which CARRY only	21	0		
ICESTORM_RAM (EBR)	5	0	30	16%
SB_IO (I/O buffers)	39	39	39	100%
ICESTORM_SPRAM	0	0	4	0%
Fmax (CLK)	62.72 MHz (PBI, PASS at 25 MHz)			

I/O budget note: the iCE40UP5K-SG48 exposes exactly 39 user I/O pins, and this design uses all 39. The mode selection (PBI vs. CCTL) is compiled into the bitstream via `VERA_IS_PBI` rather than being a runtime input, freeing the two former `DIP_SEL` pins for dedicated hardware functions: `PBI_ID_SEL` (pin 27) and `RAMBO_EN` (pin 28).

3.3 Step 3 – Place & Route

Place & route assigns logic cells and routes signals to physical iCE40 resources, guided by the pin constraints file.

```
nextpnr-ice40 --up5k --package sg48 \
    --json C6502_Interface_PBI.json \
```

```
--pcf board-pin-mapping.pcf \
--asc C6502_Interface_PBI.asc
```

Listing 4: Place & route with nextpnr-ice40 (PBI example)

The CCTL build uses the same command with `_CCTL` filename suffixes.

Important: all pin numbers in `board-pin-mapping.pcf` are placeholders. Edit the file to match your actual schematic before this step (see Section 6).

nextpnr reports timing after routing. Measured results:

Path type	Description	Worst delay
CLK → CLK	Registered path (Fmax)	15.94 ns (62.72 MHz)
async → async	Combinational (PHI2 gate)	28.12 ns
async → CLK	Input setup time	26.89 ns
CLK → async	Clock-to-output	17.17 ns

The 25 MHz system clock (period 40 ns) has **24.06 ns of slack** on the registered path. The combinational PHI2-gated outputs settle in at most 28.12 ns after address/control inputs change — well within the 6502 PHI2-high window of ≈ 279 ns at 1.79 MHz.

3.4 Step 4 – Pack the bitstream

`icepack` converts the ASCII intermediate format produced by nextpnr into the binary bitstream that is written to the SPI flash.

```
icepack C6502_Interface_PBI.asc C6502_Interface_PBI.bin
icepack C6502_Interface_CCTL.asc C6502_Interface_CCTL.bin
```

Listing 5: Pack binary bitstreams

The `.bin` file is what gets programmed into the SPI flash. Bitstream size is device-dependent (not design-dependent) on the iCE40: both bitstreams measure ≈ 104 KB (iCE40UP5K SG48).

3.5 Step 5 – Program the SPI flash

The iCE40UP5K loads its configuration from an external SPI NOR flash at power-up. The programmer writes the bitstream into that flash.

3.5.1 Using openFPGALoader (recommended)

```
# PBI bitstream (PBI connector, $D1xx)
openFPGALoader -b ice40_generic C6502_Interface_PBI.bin
# CCTL bitstream (cartridge port, $D5xx)
openFPGALoader -b ice40_generic C6502_Interface_CCTL.bin
```

Listing 6: Program with openFPGALoader

Select the programmer cable explicitly if needed:

```
openFPGALoader --cable ft2232 C6502_Interface_PBI.bin # FTDI
FT2232H
openFPGALoader --cable ft232 C6502_Interface_PBI.bin # FTDI
FT232H
openFPGALoader --cable tigard C6502_Interface_PBI.bin # Tigard
```

```
openFPGALoader --cable raspberrypi C6502_Interface_PBI.bin #
Raspberry Pi GPIO
```

3.5.2 Using icprog (FTDI FT2232H)

```
icprog C6502_Interface_PBI.bin # PBI bitstream
icprog C6502_Interface_CCTL.bin # CCTL bitstream

# Erase + write (useful if previous write was partial)
icprog -e 1048576 C6502_Interface_PBI.bin
```

Listing 7: Program with icprog

3.6 One-command build and flash

The firmware/FPGA/Makefile wraps all steps:

```
cd firmware/FPGA
make pbi # synthesise + P&R + pack ->
C6502_Interface_PBI.bin
make cctl # synthesise + P&R + pack ->
C6502_Interface_CCTL.bin
make # build both (default)
make flash-pbi # write PBI bitstream via openFPGALoader
make flash-cctl # write CCTL bitstream via openFPGALoader
make flash-pbi-icprog # write PBI via icprog (FTDI FT2232H)
make flash-cctl-icprog # write CCTL via icprog
make doc # regenerate PDF documentation
make clean # remove all generated files
```

4 Programmer Wiring

The SPI flash is shared between the FPGA (configuration bus at boot) and the programmer (writes the bitstream). During programming, the FPGA is held in reset by asserting **CRESET_B** low.

Programmer signal	iCE40UP5K pin	SPI flash pin
MOSI	SPI_SI	DI
MISO	SPI_SO	DO
SCK	SPI_SCK	CLK
CS#	–	CS#
CRST#	CRESET_B	–
CDONE	CDONE	–

Recommended SPI flash: Winbond W25Q16 (2 MB) or W25Q32 (4 MB).

5 Reset Sequencing

The VERA RBL-XE module requires a defined power-on boot sequence: the Atari computer and the VERA graphics chip must remain in reset until all FPGA configurations have completed. This is implemented with a single 74LVC1G08 logic gate — no MCU or firmware is needed.

5.1 Boot sequence

1. **Power-on:** both VERA_RESET# and ATARI_RESET# are held low.
2. **FPGA ready** (FPGA_CDONE goes high, ≈ 50 – 100 ms after power-on): VERA_RESET# is released, allowing VERA to start loading its own bitstream from its SPI flash.
3. **VERA ready** (VERA_CDONE goes high, ≈ 200 – 500 ms later): ATARI_RESET# is released and the Atari boots normally.

The natural SPI flash loading times provide all necessary delays; no RC timer or additional delay circuit is required.

5.2 Logic

Signal	Boolean expression
VERA_RESET#	FPGA_CDONE
ATARI_RESET#	FPGA_CDONE AND VERA_CDONE

FPGA_CDONE	VERA_CDONE	VERA_RST#	ATARI_RST#	Phase
0	0	0	0	Power-on / FPGA loading
1	0	1	0	FPGA ready, VERA loading
1	1	1	1	System ready

5.3 Circuit – 74LVC1G08 (SOT-23-5)

A single **74LVC1G08** (2-input AND gate, SOT-23-5) implements ATARI_RESET#. VERA_RESET# connects directly from FPGA_CDONE through a $33\ \Omega$ series resistor.

```

VCC --[4k7]--+----- FPGA_CDONE (open-drain)
              |
              +--[33R]- VERA_RESET#
              |
              +-- A -\
                    +---- 74LVC1G08 ---[33R]--- ATARI_RESET#
              +-- B -/
              |
VCC --[4k7]--+----- VERA_CDONE (open-drain)

```

Both CDONE outputs are open-drain; they require an external pullup to VCC. The $33\ \Omega$ series resistors limit current if an external driver (e.g. the Atari RESET button, which pulls the line to GND via an open-collector circuit) conflicts with the 74LVC output.

Why 74LVC and not 74HC? The 74LVC family has *5 V tolerant inputs* while operating at 3.3 V VCC. The CDONE signals are at 3.3 V; the Atari RESET line belongs to a 5 V TTL system. The 74LVC output at 3.3 V is correctly recognised as HIGH by 5 V TTL logic ($V_{IH} = 2.0\text{ V}$). The 74HC family is not suitable for this mixed-voltage interface.

5.4 Bill of materials

Ref	Value	Package	Function
U1	74LVC1G08	SOT-23-5	2-input AND gate
R1	4.7k Ω	0402	Pullup for FPGA_CDONE
R2	4.7k Ω	0402	Pullup for VERA_CDONE
R3	33 Ω	0402	Series resistor on VERA_RESET#
R4	33 Ω	0402	Series resistor on ATARI_RESET#

Total cost under 0.30 EUR; no firmware required.

6 Pin Assignment

Edit `board-pin-mapping.pcf` and replace every pin number with the actual pad number from your schematic before running `nextpnr-ice40`.

The CLK signal must be assigned to a **Global Buffer (GB) input** pin. Valid GB pins on the iCE40UP5K SG48 package are: **20, 28, 31, 38, 41, 43, 48**.

Valid user I/O pin numbers for SG48 (39 total):

2, 3, 4, 6, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21,
23, 25, 26, 27, 28, 31, 32, 34, 35, 36, 37, 38, 39,
40, 41, 42, 43, 44, 45, 46, 47, 48

Pins 1, 5, 7, 8, 22, 24, 29, 30, 33 are power/ground or reserved and **must not** appear in the PCF file — `nextpnr` rejects them with an error.

6.1 Complete pin table

Signal	Dir	PCF pin	Std	Description
CLK	In	35*	LVC MOS33	25 MHz system clock. Must land on a GB pin.
A[0]	In	40*	LVC MOS33	6502 address bit 0
A[1]	In	41*	LVC MOS33	6502 address bit 1
A[2]	In	42*	LVC MOS33	6502 address bit 2
A[3]	In	43*	LVC MOS33	6502 address bit 3
A[4]	In	44*	LVC MOS33	6502 address bit 4
A[5]	In	45*	LVC MOS33	6502 address bit 5
A[6]	In	46*	LVC MOS33	6502 address bit 6
A[7]	In	47*	LVC MOS33	6502 address bit 7
A[8]	In	48*	LVC MOS33	6502 address bit 8
A[9]	In	6*	LVC MOS33	6502 address bit 9
A[10]	In	2*	LVC MOS33	6502 address bit 10
A[11]	In	3*	LVC MOS33	6502 address bit 11
A[12]	In	4*	LVC MOS33	6502 address bit 12
A[13]	In	26*	LVC MOS33	6502 address bit 13. Also wired to SRAM A[13] on PCB.
A[14]	In	9*	LVC MOS33	6502 address bit 14
A[15]	In	10*	LVC MOS33	6502 address bit 15

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Signal	Dir	PCF pin	Std	Description
D[0]	InOut	11*	LVC MOS33	6502 data bit 0. Shared net with SRAM DQ[0].
D[1]	InOut	12*	LVC MOS33	6502 data bit 1. Shared net with SRAM DQ[1].
D[2]	InOut	13*	LVC MOS33	6502 data bit 2. Shared net with SRAM DQ[2].
D[3]	InOut	14*	LVC MOS33	6502 data bit 3. Shared net with SRAM DQ[3].
D[4]	InOut	15*	LVC MOS33	6502 data bit 4. Shared net with SRAM DQ[4].
D[5]	InOut	16*	LVC MOS33	6502 data bit 5. Shared net with SRAM DQ[5].
D[6]	InOut	17*	LVC MOS33	6502 data bit 6. Shared net with SRAM DQ[6].
D[7]	InOut	18*	LVC MOS33	6502 data bit 7. Shared net with SRAM DQ[7].
PHI2	In	34*	LVC MOS33	6502 phase-2 clock. Synchronised internally; not used as FPGA clock.
RNW	In	36*	LVC MOS33	Read/Not-Write. High = read, Low = write.
VERA_CS	Out	32*	LVC MOS33	VERA chip-select latch output (active low).
MPD	Out	31*	LVC MOS33	Machine Program Data (active low, PBI mode only). Asserted during PBI ROM (\$D800–\$DFFF) reads; blocks Atari from driving D[7:0]. Always inactive (high) in CCTL mode.
EXTSEL	Out	37*	LVC MOS33	External Select (active low). PBI mode: asserted for PBI RAM (\$D600–\$D7FF) and RAMbo (\$4000–\$7FFF). CCTL mode: asserted for RAMbo only. Blocks Atari internal RAM from responding.
PBI_ID_SEL	In	27*	LVC MOS33	Hardware jumper to one of D[0..7] on PCB. Sampled when \$D1FF is written (PBI mode only); <code>vera_cs_latch <= not PBI_ID_SEL</code> . Logic 1 on the chosen bit selects VERA, logic 0 deselects. Ignored in CCTL build.
RAMBO_EN	In	28*	LVC MOS33	Tie to VCC (pullup) to enable RAMbo hardware; tie to GND (pulldown) to disable RAMbo completely, overriding PBCTL[2]. Both RAMBO_EN and PBCTL[2] must be high for RAMbo to respond.
SRAM_A_BANK[0]	Out	19*	LVC MOS33	SRAM address bit 14 (bank LSB).
SRAM_A_BANK[1]	Out	20*	LVC MOS33	SRAM address bit 15.
SRAM_A_BANK[2]	Out	21*	LVC MOS33	SRAM address bit 16.
SRAM_A_BANK[3]	Out	38*	LVC MOS33	SRAM address bit 17 (bank MSB).
SRAM_CE_N	Out	23*	LVC MOS33	SRAM chip enable (active low). Asserted during PHI2-high when address is in \$4000–\$7FFF and RAMbo is enabled.

continued on next page

Signal	Dir	PCF pin	Std	Description
SRAM_OE_N	Out	39*	LVC MOS33	SRAM output enable (active low). Asserted on read cycles only.
SRAM_WE_N	Out	25*	LVC MOS33	SRAM write enable (active low). Asserted on write cycles only. Pulse width equals PHI2-high time (≈ 279 ns at 1.79 MHz).

* Pin numbers are placeholders. Replace with actual pad numbers from your board schematic.

6.2 External SRAM address wiring

The SRAM address bus is split between PCB traces (lower 14 bits) and FPGA outputs (upper 4 bank bits):

SRAM address bit	Source	Note
A[13:0]	6502 A[13:0] (PCB net)	Direct trace, no FPGA I/O pin
A[14]	FPGA SRAM_A_BANK[0]	Bank bit 0 = PORTB[2]
A[15]	FPGA SRAM_A_BANK[1]	Bank bit 1 = PORTB[3]
A[16]	FPGA SRAM_A_BANK[2]	Bank bit 2 = PORTB[5]
A[17]	FPGA SRAM_A_BANK[3]	Bank bit 3 = PORTB[6]
A[18]	GND (PCB)	Selects lower 256 KB of 512 KB chip

6.3 RAMbo bank formula

The FPGA snoops writes to the Atari PIA to derive the active bank:

```
PORTB at $D301  -- bank bits
PBCTL at $D303  -- bit 2 enables RAMbo

bank[1:0] = PORTB[3:2]
bank[3:2] = PORTB[6:5]
=> SRAM A[17:14] = { PORTB[6], PORTB[5], PORTB[3], PORTB[2] }
```

RAMbo is active only when both RAMBO_EN (pin 28 tied to VCC) and PBCTL[2] = 1 (software enable written by the Atari OS).

7 Timing

7.1 Read cycle (SRAM)

Event	Time from PHI2 rise	Note
Address valid (6502 t _{ADS})	−40 ns	Before PHI2 rise
SRAM_CE# asserted	≈ 0 ns	Combinational gate on PHI2
SRAM data valid (t _{AA} =55 ns)	+55 ns	AS6C4008-55
6502 data setup deadline	+249 ns	279 ns − 30 ns t _{DSR}
Read margin	+194 ns	Adequate

7.2 Write cycle (SRAM)

$WE\#$ is asserted combinationally when PHI2 is high and the address is in the RAMbo window. The pulse width equals the PHI2 -high time:

$$t_{WC} = t_{\text{PHI2-high}} \approx 279 \text{ ns} \gg t_{WC(\text{SRAM})} = 55 \text{ ns}$$

7.3 PBI RAM / ROM (BRAM)

BRAM reads are triggered on `phi2_rise` (one clock cycle after the real PHI2 edge, $\approx 40 \text{ ns}$ delay). The BRAM output is registered and valid within one additional clock cycle ($\approx 40 \text{ ns}$), leaving $\approx 199 \text{ ns}$ of margin before the 6502 samples the bus.

8 Source File Reference

File	Description
<code>C6502_Interface.vhd</code>	Top-level VHDL entity (Rev 3). Generic <code>VERA_IS_PBI</code> selects PBI or CCTL mode at synthesis time. Contains all bus logic, BRAM instances, and SRAM control.
<code>custom_types.vhd</code>	Package defining <code>rom_array</code> (2048×8) and <code>ram_512_array</code> (512×8) array types.
<code>pbi_rom_pkg.vhd</code>	Auto-generated package containing the <code>PBI_ROM_INIT</code> constant (2048 bytes). Do not edit manually.
<code>gen_pbi_rom_pkg.py</code>	Python 3 script that converts a 6502 <code>.bin</code> file into <code>pbi_rom_pkg.vhd</code> . Invoked by the MCU Makefile.
<code>board-pin-mapping.pcf</code>	Pin constraints for iCE40UP5K SG48. All pin numbers are placeholders – edit before use.
<code>Makefile</code>	Targets: <code>pbi</code> , <code>cctl</code> , <code>all</code> , <code>flash-pbi</code> , <code>flash-cctl</code> , <code>flash-pbi-iceprog</code> , <code>flash-cctl-iceprog</code> , <code>doc</code> , <code>clean</code> .